

Bound Low-Side Drive Intent

Make every requested microamp earn admission

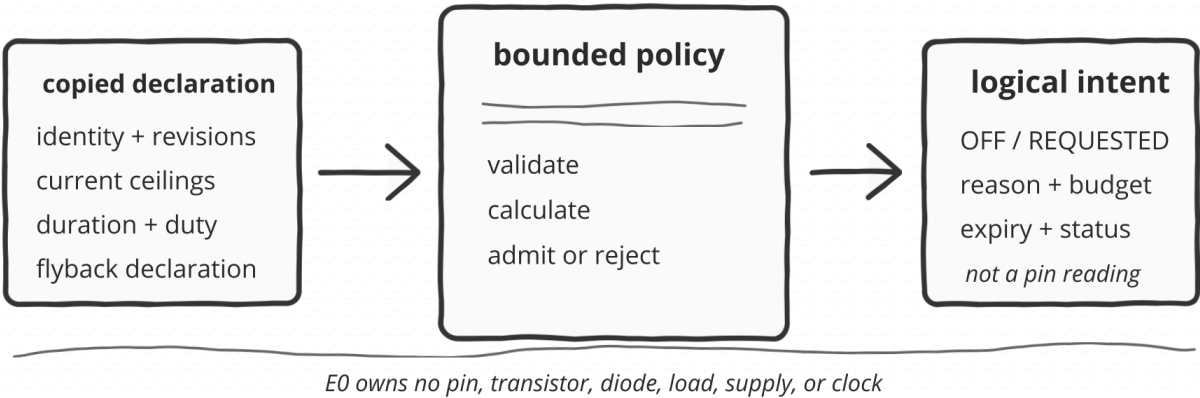
E0 COPIED-DECLARATION REPLAY — NO PIN, POWER, OR SWITCHING CLAIM

Lesson	079 — component	Time	150–210 minutes
Prerequisite	fixed-point arithmetic and supplied time	Board	compile-only Mega replay
Input	one declaration and copied requests	Observation	named intent cells

Responsibility. `BoundedLowSideDriverPolicy` validates one immutable declaration, performs checked conservative current and duty arithmetic, and publishes bounded active-high or all-off logical intent. It owns no GPIO, endpoint, transistor, diode, load, supply, timer, clock, or storage.

1 Start with authority, not a transistor picture

This lesson deliberately contains no circuit or schematic. A familiar part number does not fix pin order, gain, package, voltage, current, carrier topology, or whether the populated device is genuine. PN2222 and S8050 remain separate variants. The E0 policy can check copied declarations; it cannot identify or energize a physical specimen.



Alternative text: a rough graphite declaration card flows through a bounded software policy into a logical intent card; a heavy underline says the E0 boundary owns no physical hardware or clock.

E0 establishes	E0 leaves open
declaration structure and correlation	exact transistor, diode, load, and pinout
checked current-budget arithmetic	voltage, current, switching, and temperature
bounded active-high logical intent	GPIO ownership and observed output level
deterministic off/fault intent	independent power removal and safe-state proof

2 Declare one topology precisely

The sole modeled topology is an active-high bare-NPN low-side drive. **outputLevelHigh** means the logical level a future endpoint would be asked to produce; it is never a pin observation. There is no active-low, high-side, PNP, MOSFET, Darlington, H-bridge, relay-module, or generic three-pin interpretation.

A resistive-indicator declaration requires flyback to be not required and every diode field to be canonically absent. An inductive-inert declaration requires a present diode plus nonzero identity, revision, orientation, return, reverse-voltage, and forward-current fields. E0 checks completeness and conservative rating arithmetic only; it does not authenticate placement or observe clamping.

Declaration	Structurally valid?	Physical authority?
resistive, diode fields all absent	yes	none
resistive with a stray diode identity	no	none
inductive with one missing diode field	no	none
complete copied inductive declaration	possibly	still none

Prediction: does a complete inductive declaration prove that a diode is installed in the correct direction? _____. Explain: .

3 Bind the whole declaration

Schema, specimen family and specimen identity, revisions, evidence source, configuration, topology, protection, every current/voltage bound, resistor tolerance, maximum duration, duty window, and duty limit enter one canonical descriptor identity digest. Zero is a valid digest value, not a missing-value sentinel.

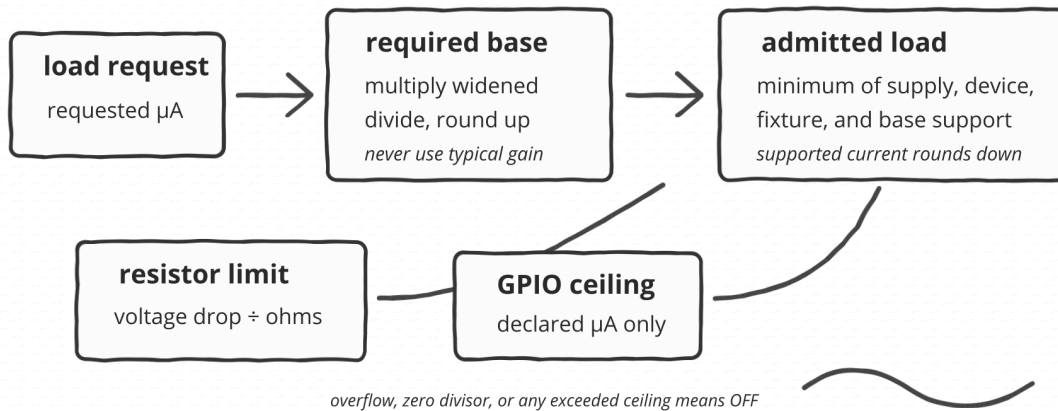
An accepted request additionally binds lifecycle generation, session, run, step, request, source, source-configuration revision, sequence, observation time, duration, active state, requested current, and complete producer status. Identity drift, changed duplicates, gaps, regressions, or half-range ambiguity reject atomically.

4 Do conservative current arithmetic

All current values are integer microamps, voltage values are integer millivolts, and resistance is integer ohms. Multiplications widen before use. Required base current rounds up; load current supported by admitted base current rounds down.

$$I_{\text{base,req}} = \left\lceil \frac{I_{\text{load,req}} G_d}{G_n} \right\rceil \quad I_{\text{base,res}} = \left\lfloor \frac{\max(0, V_{\text{OH,min}} - V_{\text{BE,max}}) 1000}{R_B} \right\rfloor$$

$$I_{\text{load,adm}} = \min(I_{\text{load,req}}, I_{\text{supply,avail}}, I_{\text{device}}, I_{\text{fixture}}, \lfloor I_{\text{base,adm}} G_n / G_d \rfloor)$$



Alternative text: a hand-drawn load request passes through an upward-rounded required-base calculation into a downward-rounded admitted load calculation; resistor, GPIO, supply, device, and fixture ceilings all feed the decision.

1. With requested load $20,001 \mu\text{A}$, gain numerator 10, and denominator 1, required base current is _____.
2. Available supply is supply limit minus reserved supply. If reserved exceeds the limit, should arithmetic saturate to zero? _____.
3. If every ceiling passes except fixture load, the result is _____.

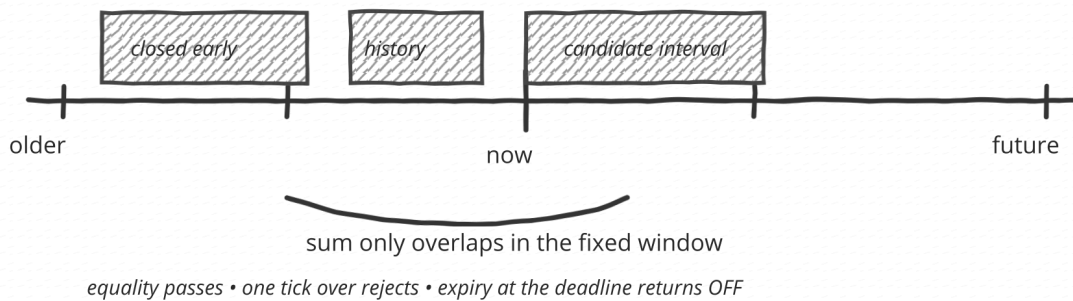
Zero divisors or required ceilings, overflow, underflow, an unrepresentable result, and any exceeded binding ceiling never become a plausible partial command. They reject without mutation and publish no hardware fact.

5 Bound duration and rolling duty

An active request supplies a nonzero duration no greater than the configured maximum. Admission freezes **expiresAt**; replay never extends it. **update(now)** receives time from the caller and returns to canonical off at deadline equality or later. The policy reads no clock.

Eight fixed history entries retain admitted intervals. Early off or cancellation shortens the current interval, but the used interval remains in the rolling duty history. Old intervals are pruned only when their ends lie at or before the window start.

rolling duty window at supplied “now”



Alternative text: a graphite time line shows an early-closed interval, retained history, supplied now, and a candidate interval; a brace marks that only rolling-window overlaps count toward duty.

The widened admission inequality is

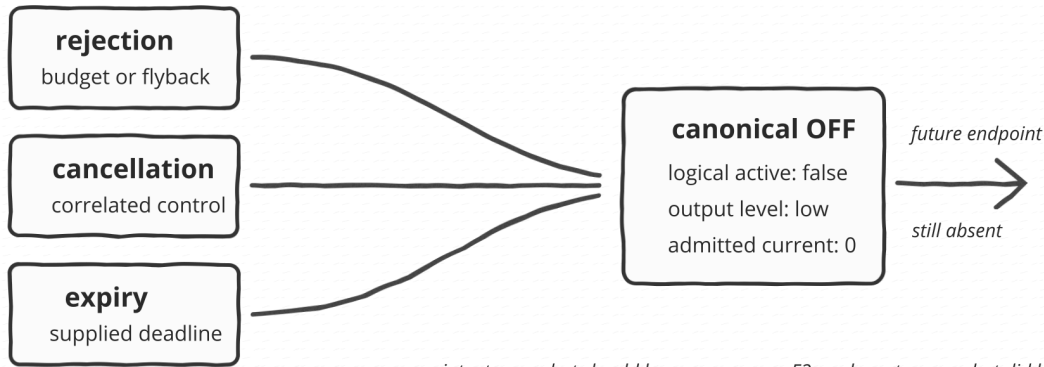
$$\text{reservedTicks} \times 1000 \leq \text{windowTicks} \times \text{maximumDutyPer mille}.$$

Equality passes; one tick or permille over rejects. A ninth still-overlapping interval reports **Rejected/CapacityExceeded**, preserves all eight history entries, and publishes off intent for that request. This finite boundary is a domain result, not hidden allocation.

Case	Expected interpretation
deadline minus one tick	request may remain active
deadline exactly	Expired , canonical off
duty equality	admitted if every other check passes
one unit over duty	rejected, history remains attributable

6 Make all-off a complete result

Logical inactive requests, rejection, cancellation, producer fault, expiry, reset, and shutdown all publish active-high topology's canonical off intent: `logicalActive=false`, `outputLevelHigh=false`, and zero admitted currents. The reason and producer status remain attributable.



intent says what should happen; an open E2 card must prove what did happen

Alternative text: sketched rejection, cancellation, and expiry cards converge on one canonical off card with false logical activity, low output intent, and zero admitted current; the future physical endpoint is shown as absent.

Cancellation uses a fully correlated control value. If off is confirmed, the terminal result is **Cancelled**. If an attributable producer failure prevents confirming off, the result is **Fault**; cancellation remains the cause and cannot hide failed safe return. A well-formed producer fault is domain evidence and may return operation status OK while intent remains off.

Atomicity experiment. Place recognizable canaries around the output intent. Submit a changed duplicate, a sequence gap, a backward time, and a malformed request. Confirm that every API rejection leaves the policy, output, and canaries byte-for-byte unchanged. Then contrast a valid budget rejection, which commits an attributable off/rejected domain result.

7 Replay without touching hardware

The Mega sketch constructs one copied descriptor, begins one session, applies bounded active and off requests, exercises one rejection and expiry, and copies results into named volatile cells.

1. **Predict** state, reason, output level, required base current, admitted load current, and expiry before each call.
2. **Observe** the named volatile intent cells with a debugger, simulator, or disassembly. Serial is optional and never sole evidence.
3. **Interpret** agreement as deterministic E0 policy evidence. It does not show that a physical pin changed or a load drew current.

8 Diagnose by precedence

Symptom	Inspect first	Interpretation
non-OK API status	structure, lifecycle, correlation	state and output stay unchanged
SourceIneligible	copied source packet and eligibility	powered work remains prohibited
BaseBudgetInsufficient	resistor, voltage, GPIO, gain ratio	conservative base support is too small
FlybackMissing	energy and all diode declarations	inductive intent has no complete protection declaration
BudgetExceeded	supply, device, fixture ceilings	requested load exceeds a binding bound
Expired	supplied time and frozen deadline	replay did not extend authority
CapacityExceeded	eight overlapping reservations	fixed history is intentionally full
Fault during cancel	off confirmation and status	cancellation cannot mask failed safe return

9 Test the complete contract

- Exhaust enum encodings, load/flyback cross-products, canonical absence, all identity/revision mismatches, and one-field digest mutations.
- Exercise zero divisors and ceilings, rounding vectors, widened multiplication boundaries, exact/one-over budgets, and output canaries.
- Test duration, expiry, early close, eight-entry pruning, ninth-entry exhaustion, duty equality/one-over, rollover, and half-range time.
- Test inactive/active intent, source ineligibility, producer fault, cancellation/fault collision, reset, shutdown, and idempotence.

10 Open acceptance cards

E0 software	Closed: strict host tests and sanitizer replay, headers/style, Mega compile, exact resource evidence, independent disposition, lesson, publication, and terminal architecture review pass.
E1 identity	Open: both-face photographs, markings, primary sources, traced pin order/topology, passive measurements, energy classification, and reviewed specimen descriptor. No powered observation belongs to E1.
E2b fixture	Open: exact transistor, diode, and load; GPIO/resource and rail map; base limiting; current-limited supply; authoritative schematic; test points; stored-energy disposition; rollback; independent power removal; and recorded bench acceptance.
Physical result	Open. This lesson contains no circuit, schematic, powered observation, safe-state measurement, or supported-specimen claim.

11 Read the measured software boundary

The canonical repository Mega replay measures 11,560 B flash and 561 B static SRAM ordinarily. The isolated exact no-LTO probe records:

Metric	Actual	Target	Hard	Disposition
flash	12,974 B	10,240 B	14,336 B	reviewed miss
static SRAM	561 B	768 B	1,024 B	pass
synchronous stack	320 B	320 B	448 B	pass at equality
policy object	240 B	192 B	256 B	reviewed miss
descriptor object	96 B	96 B	128 B	pass at equality
residual SRAM	7,183 B	4,096 B	3,072 B	pass

The flash and policy target misses are independently accepted below their hard limits; every other target and every hard limit pass. The exact result is bound to fingerprint 4972bd9733d608d52ac81bfb1320e61088b10c3e59910f3be8c439b5838d33e7. These measurements prove the software boundary and no physical behavior.

Acceptance record — leave open until observed.

Exact specimen and revision: _____
 Primary-source packet and reviewer: _____
 Fixture, current limit, and independent power removal: _____
 Resource-acquisition observation: _____
 All-off electrical observation and named test point: _____
 Bench date, result, and reviewer: _____

12 Exercises

1. Explain why “output level low” is intent, not safe-state evidence.
2. Construct a current request where upward and downward rounding each change the answer by one microamp.
3. Explain why early cancellation shortens the active interval but cannot erase it from duty history.
4. List the additional evidence required before drawing an authoritative schematic for one exact E2b fixture.